



Design and performance evaluation of a new cyclone separator



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ABSTRACT

This paper presents the design and performance characteristics of a new type cyclone separator. The design of this cyclone is based on the idea of improving cyclone efficiency by increasing the vortex length. This cyclone differs from a classical cyclone with the separation space. Instead of conical part, the separation space of this cyclone consists of an outer cylinder and a vortex limiter. A model of this cyclone has been manufactured and tested. Collection efficiency and pressure drop of this cyclone were measured under different operational conditions. Experimental results were compared with the conventional cyclones. The effects of the position of the vortex limiter on the cyclone performance were also investigated.

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1. Introduction

Separation, collection and sampling aerosol particles or droplets from the carrying flow is an important topic in the field of ambient sampling and industrial particulate control. Cyclone is one of the most widely used devices for the separation process because of its simple design and lower construction and operating costs. There are many geometric and operational parameters influenced the separation process and cyclone performance which mainly comprises collection efficiency and pressure drop. Starting with Alexander (1949), many studies have been conducted to improve cyclone performance by evaluating the effects of geometric and operational parameters.

Modifications to the inlet design and the effects of inlet geometry on the cyclone performance have been studied extensively. The effect of the cyclone inlet dimensions on the performance and flow field pattern has been investigated computationally by Elsayed & Lacor (2011). They found that the effect of changing the inlet width is more significant than the inlet height especially for collection efficiency. The inlet area is one of the relevant parameters influencing the natural vortex length (Alexander, 1949; Cortés & Gil, 2007; Avci et al., 2013). Zhao et al. (2006) compared the performance of two types of cyclones with the conventional single inlet and spiral double inlets. The numerical results show that, the new type cyclone separator using the adding spiral double inlet can improve the symmetry of gas flow pattern and enhance the particle separation efficiency. The effect of inlet section angle has been tested by a few researchers (Qian & Zhang, 2007; Qian & Wu, 2009). Also Erdal & Shirazi (2006) investigated the effect of three different inlet geometries (one inclined inlet, two inclined inlets and a gradually reduced inlet nozzle) on the flow behavior. Three cyclone separators with different inlet geometry namely, conventional tangential single inlet, direct symmetrical spiral inlet, and a converging symmetrical spiral inlet, were investigated and compared as a function of particle size and flow rate in the work by Zhao et al. (2004).

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The effects of cone dimension on the flow pattern and cyclone performance were also investigated in the literature (Xiang & Lee, 2001; Chuah et al., 2006; Xiang et al., 2011). Researchers showed that when the cone opening is larger than the gas outlet diameter, reduction in cone size results in higher collection efficiency without significantly increasing the pressure drop. In the study of Yoshida et al. (2001), various types of apex cones were used at the inlet part of the dust box. They found that the effect of the apex cone angle on the particle separation performance decreases at high inlet velocity conditions. Effects of a cone prolonged with a vertical tube (dipleg) on the separation performance of a cyclone were also studied (Qian et al., 2006; Kaya & Karagoz, 2009). The effects of a counter-cone in the bottom of the cyclone on the cyclone performance have also been investigated (Yoshida et al., 2003, 2010; Kepa, 2012).

The effects of the shape and diameter of the vortex finder on the cyclone's performance were also studied by many researchers. Xiang & Lee (2008) showed that the tangential velocity increases as the exit tube diameter is reduced, giving rise to higher particle collection efficiency, and the pressure drop decreases with the increasing exit tube diameter. Simulations on the separation efficiency and pressure drop of a modified cyclone separator with a novel exhaust at different exhaust insert depths, sloping orientations, contracting angles and sloping angles were performed by Chen & Liu (2010). Kepa (2010) analyzed numerically the flow inside a cyclone with the division of gas and particle flow in a vortex finder in which an additional internal tube (an inner vortex finder) was placed inside the vortex finder of a traditional cyclone.

Additionally, more parameters were also considered in the literature. For example, cyclone height was studied by Hoffman et al. (2001) and Safikhani et al. (2010). Effects of a stick inserted in a cyclone were studied by Wang et al. (2005) and Farahani et al. (2011), performance of square cyclones were investigated by Su & Mao (2006), Safikhani et al. (2011) and Su et al. (2011). Kim et al. (2001) worked on the cyclone with different body designs including the conventional smooth surface body, the spiral guide body, the circumferential groove body and the vertical groove body. The results showed that the spiral guide can help enhance collection efficiency at low flow rates. However, at high flow rates, the guide no longer plays an important role in cyclone collection efficiency.

Optimization of dimensions of a conventional cyclone was also taken attention of researchers. Optimization of the cyclone geometry for minimum pressure drop using mathematical models and CFD simulations was performed by Elsayed & Lacor (2010, 2012). A new set of geometric ratios which are very close to the Stairmand high efficiency design was obtained to achieve minimum pressure drop.

On the other hand, different cyclone designs have been reported in the literature. One of them is the double cyclone which was proposed and investigated by Zhu et al. (2001), Xiang & Lee (2001) and Lim et al. (2004). They evaluated its performance and compared to the high efficiency Stairmand cyclone. They found that the double cyclone has a lower pressure drop but has not higher collection efficiency than a conventional cyclone with the same overall dimensions. However, the collection efficiency of the double cyclone with electric field increased greatly with the increasing the applied voltage. Another type of cyclone technology called the circumfluent cyclone was developed and studied by Wang et al. (2010). They investigated experimentally its separation efficiency and pressure distribution, and compared its characteristics with the conventional cyclone. The results showed that the collection efficiency of this cyclone is higher by 8% than that of a conventional cyclone in the gas inlet velocity range of 12–26 m/s and the pressure drop within this cyclone is only one half or one third of that of the conventional cyclone.

Experimental and simulated studies of Yoshida et al. (2001) were conducted on the particle classification performance of Iinoya's type cyclone which has double conical sections. They found that separation efficiency of conventional and Iinoya's type cyclones are nearly equal; however, pressure drop of the Iinoya's type cyclone is about 10% smaller than that of the conventional cyclone. Alternatively, an auxiliary device named Post Cyclone was developed to reduce the emission of fines particles from the exhaust of reverse flow cyclones (Jo et al., 2000). The Post cyclone consists of two annular shells placed vertically on the exit of a conventional cyclone. The experiments demonstrated that the Post Cyclone increases the overall efficiency of the cyclone by 2–20% depending on the operating conditions and the size of the cyclone. The pressure drop over the Post Cyclone is approximately 10% of the total pressure drop over the cyclone.

The aim of the present work is to present the design and performance evaluation of a new cyclone separator. This cyclone differs from a classical cyclone with the separation space. Instead of conical part, the separation space of this special cyclone consists of an outer cylinder, and a plate vortex limiter to end the vortex. The movable vortex limiter allows us to adjust the vortex length according the inlet velocity in order to attain high efficiency.

2. Design

There are a wide variety of cyclone separators with different shapes and configurations. The most commonly used one is the reversed flow cyclone which has a cone under a cylindrical body with a tangential inlet. Separation performance strongly depends on the structure and length of the double vortex formed in this kind of cyclone. According to the mathematical model explained in the works by Avci and Karagoz, friction factor which is a function of surface roughness and the Reynolds number, plays an important role on the cyclone efficiency and pressure drop (Avci & Karagoz, 2003; Karagoz & Avci, 2005). It was also demonstrated that surface roughness affects axial and tangential velocity distributions, and consequently the vortex length, in a cyclone (Kaya et al., 2011). These results have motivated the idea that the vortex length and separation performance can be increased by decreasing friction losses in a cyclone. The design of this new cyclone is based on the idea that the vortex length and separation performance can be increased by decreasing friction losses in a cyclone. The new design differs from a classical cyclone with the separation space. It has inner and outer

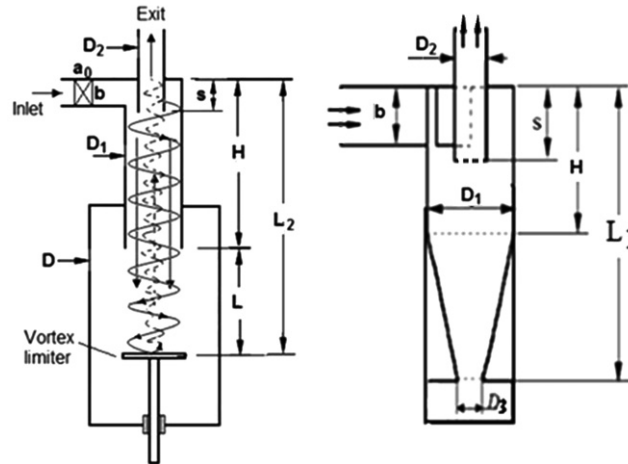


Fig. 1. New design and conventional cyclones.

Table 1
Dimensions of the cyclones tested in this study.

Cyclone dimensions	Size (mm)
Body diameters	$D_1 = 190$, $D = 250$.
Vortex finder diameter, D_2	100
Inlet section ($a_0 \times b$)	38×95
Cylinder height, H	490, 885
Cyclone length, L_1	970, 1365
Position of the vortex limiter, L	320, 480, 640, 800
Vortex finder insertion length, S	$S = b$
Diameter of the cone tip, D_3	84

cylinders without a conical part, and a vortex limiter (Fig. 1). The flow enters tangentially into the inner cylinder which has the main frictional surface and serves as vortex creator. The flow spirals down below the inner cylinder almost without wall friction through the outer cylinder and then turns back from the limiter plate towards the exit pipe. The vortex limiter can be moved up and down unable to adjust the vortex length. For example, if the flow rate or inlet velocity increases the vortex limiter can be slide down to increase the vortex length and to attain high efficiency. The particles move towards the wall of the outer cylinder and are separated from the vortical flow under the action of centrifugal forces and collected on the bottom of the outer cylinder. That is the main separation process occurs in the outer cylinder. The new design has also the advantages of easy manufacture and low maintenance. A prototype was constructed to test its performance. The geometric dimensions of the prototype are given in Table 1.

3. Experimental setup

In order to compare the performance of the new cyclone with conventional cyclones, a conventional reversed flow cyclone which has the same body diameter, inlet and outlet sections were also constructed (Fig. 1). The conventional cyclone was tested at the same operational conditions. The experimental setup used for testing the cyclones is shown in Fig. 2. The air flow through the cyclone was supplied by a radial ventilator from the atmosphere. Pressure drop was measured with a differential manometer (TESTO 521) at each test. The particulate material was injected into the main air flow by means of an ejector. The solid particles used in the test were cement with a d_{10} , d_{50} and d_{90} of approximately 2, 14 and 48 μm , respectively. The solid loading was 15 gr/min. Samples taken from the fed and collected particles were analyzed by using a Malvern Mastersizer 2000. Experiments have been carried out for different inlet velocities ranging from 10 to 25 m/s. The length of the main separation space (L) can be adjusted by changing the position of the vortex limiter.

4. Results and discussion

4.1. Comparison of the cyclone performances

A set of experiments were performed in order to measure the pressure drop and collection efficiency of the cyclones. First of all, the new cyclone with $L = 480$ mm and $H = 490$ mm was tested and the results were compared with the results

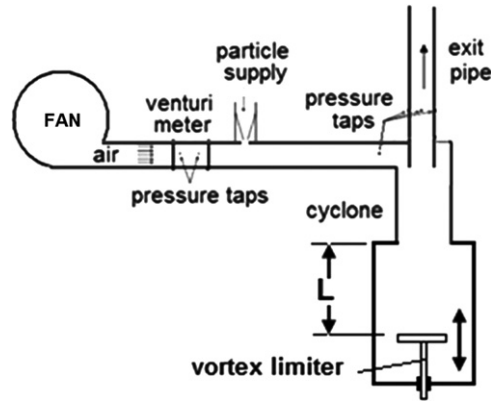


Fig. 2. Schematic representation of experimental set-up.

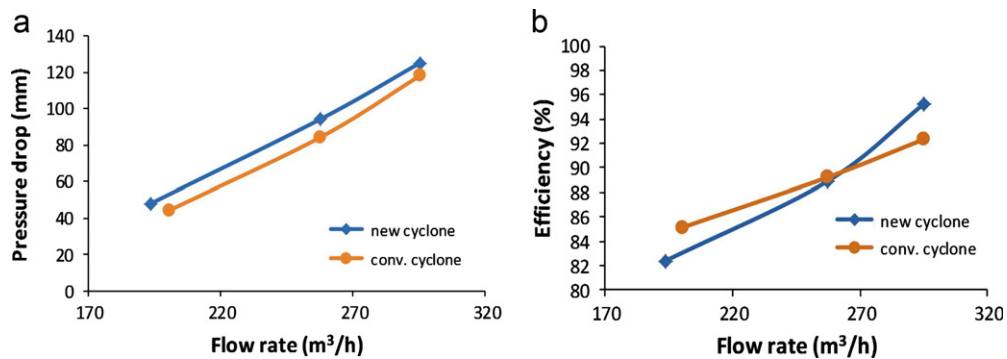


Fig. 3. Comparisons of pressure drops (a) and collection efficiencies (b) between the new and conventional cyclones (L was kept constant at 480 mm for the new cyclone).

obtained in the conventional cyclone with the same size of $H=490$ mm and $L_1=970$ mm (Fig. 3). The values of the inlet velocities were 15–19.5–22.5 m/s. As it can be seen from the figure, the new cyclone gives generally higher pressure drop (Fig. 3a). However, particle collection efficiency of the new cyclone is better comparing to the conventional one, especially for high flow rates (Fig. 3b). On the other hand, the position of the vortex limiter (i.e. L) is adjustable. Therefore, the collection efficiency of the new cyclone can be increased by adjusting the position of the vortex limiter according the flow rate. As it can be seen from Fig. 4, adjusting the position of the vortex limiter leads to an increase up to about 5% in efficiency of the new cyclone while pressure drop is almost the same with the conventional cyclone.

4.2. Effects of the vortex limiter on the performance of the new cyclone

Detailed analyses were carried out for the new cyclone of $H=885$ mm. Tests were repeated for four different positions of the vortex limiter ($L=320, 480, 640, 800$ mm). Fig. 5 shows pressure drop values through the new cyclone. Pressure drop curves are almost the same and differences between them are not significant. Similar results of pressure drop were obtained for $H=490$ mm.

Particle collection efficiency data obtained in these tests was summarized in Fig. 6. Collection efficiency increases with the increasing flow rate as expected and lowest efficiency was obtained for $L=320$ mm. It seems also that different behavior, which is the rapid increase in efficiency, occurs in high flow rates for the longest ($L=800$ mm) positions of the vortex limiter. This might be caused by longer vortex length due to high flow rates for this case.

Adjustment of the position of the vortex limiter makes possible to equalize the natural vortex length to the cyclone length ($H+L$) for which efficiency reaches the highest value. Fig. 7 supports this thought for the new cyclone with $H=885$ mm. The collection efficiency and the cut size (or the critical diameter, d_{cr}) which is the particle diameter collected with 50% efficiency were presented in Fig. 7. The critical diameter, d_{cr} , decreases as the length L increases. As the length increases further, the cut size is seen to increase again indicating lower collection efficiency. As a result, the cut size has a minimum at a certain value of L , depending on the flow rate. This finding shows that there exists an optimum length of L for the best collection efficiency. Similar result was obtained theoretically by Surmen et al. (2011) for conventional cyclones. This optimum cyclone length might be equal to the natural vortex length for each flow rate, so that collection efficiency will be a maximum. At a certain flow rate, there is an optimum position of the vortex limiter.

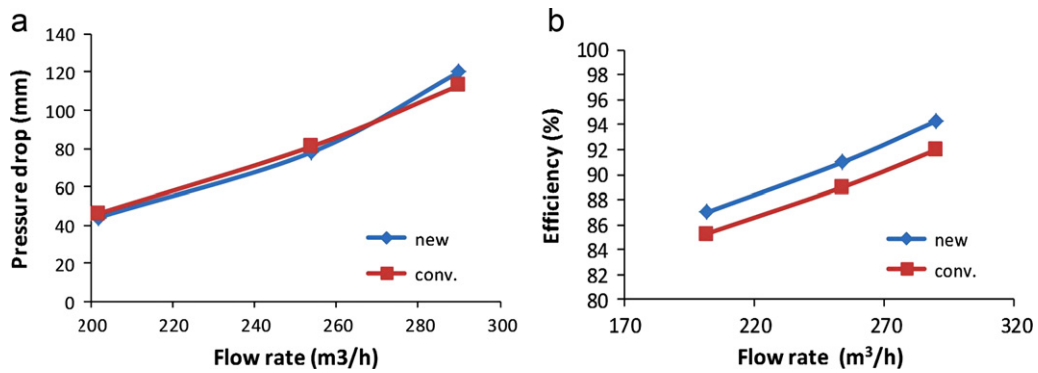


Fig. 4. Comparisons of pressure drops (a) and collection efficiencies (b) between the new and conventional cyclones (L was adjusted).

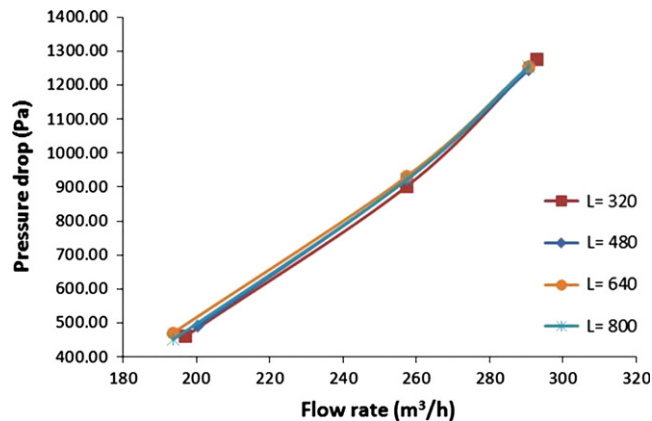


Fig. 5. Pressure drop through the new cyclone with $H=885$ mm.

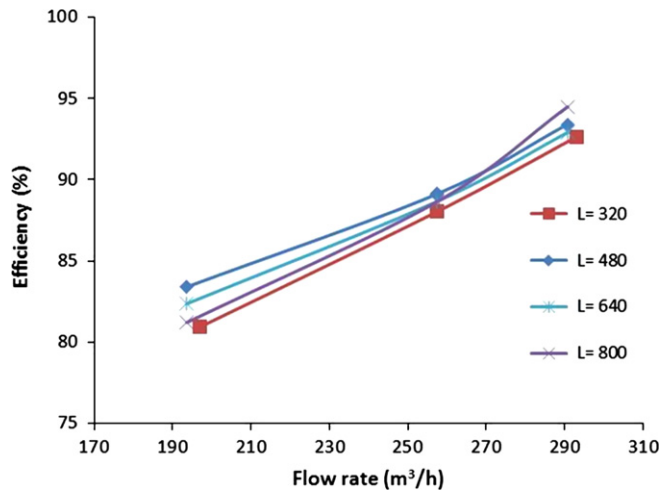


Fig. 6. Particle collection efficiency curves of the new cyclone with $H=885$ mm.

Fractional efficiency is another important performance parameter in cyclones and it depends on many parameters. Among them, the effects of L , H and flow rate were investigated in this study. Figs. 8 and 9 show the fractional efficiency curves for two different body heights ($H=490$ and 885 mm). It is possible to say from the comparison of the curves in Fig. 8 which has dimensionless x axis that the general trend of the curves remains the same. However, the efficiency curves of the new cyclone are steeper than that of the conventional cyclone. The steepness of the curves is changing as the position of the vortex limiter varies. The cyclones L320 ($L=320$ mm) and L800 ($L=800$ mm) have the steepest curves for $H=490$ mm. Fig. 9 has a dimensional x axis in order to compare efficiencies for a certain particle diameter. The best

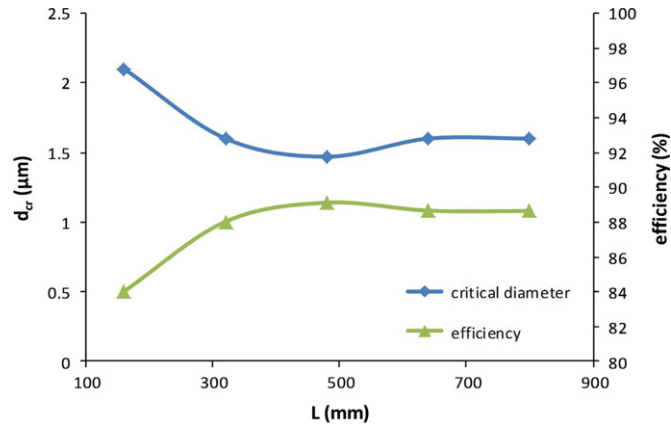


Fig. 7. Effects of L on the collection efficiency of the new cyclone with $H=885$ mm ($Q=257$ m³/h).

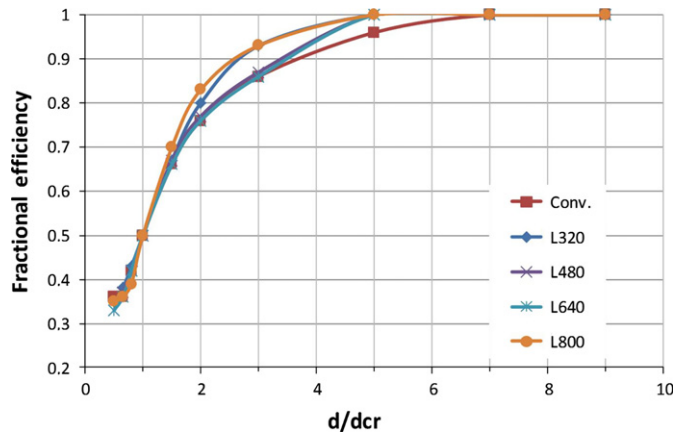


Fig. 8. Fractional efficiency curves of the new and conventional cyclones for $H=490$ mm ($Q=257$ m³/h).

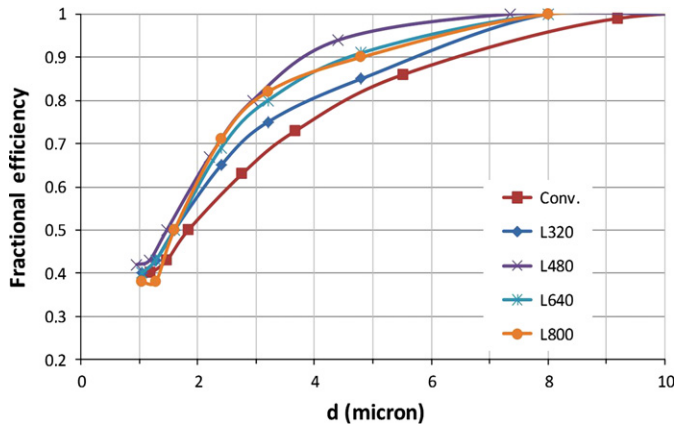


Fig. 9. Fractional efficiency curves of the new and conventional cyclones for $H=885$ mm ($Q=257$ m³/h).

collection efficiency for fine particles obtained by the cyclone with $L=480$ mm. This cyclone (L480) also has the steepest curves for $H=885$ mm.

5. Conclusion

This study introduces a novel design of tangential inlet cyclone separators. The new design cyclone has a double cylindrical body and a vortex limiter instead of a conical body seen in conventional cyclones. The performance parameters

of the new cyclone were investigated experimentally by using cement particles injected into the inlet air flow. The effects of structural difference on the cyclone performances were observed in the comparison of the results between the new and conventional cyclones. Moreover, the effects of height of the adjustable vortex limiter on the cyclone performance were clearly seen in the results of the new cyclone.

The conclusions of the present experimental study are summarized below:

1. Depending on particles characteristics, particles may accumulate on the conventional cyclone internal surfaces and lead to an increase in surface roughness. This phenomenon decreases the cyclone performance. However, this rarely or never occurs in the new cyclone.
2. The particles falling down on the internal surface may be dragged into the internal vortex which goes to exit in conventional cyclones and collection efficiency decreases. This is not the case in the new cyclone.
3. The length of the new cyclone can be increased easily, if required. However, there is no possibility to increase the length of a conventional cyclone.
4. In the new cyclone, dust collected in the dustbin can be discharged easily and continuously.
5. The duration for particle separation can be increased by increasing the cyclone length without causing extra pressure drop in the new cyclone design.
6. High collection efficiency can be obtained by adjusting the vortex limiter according the flow rate or inlet velocity.
7. There is an optimum cyclone length which leads to the best collection efficiency for a given flow rate in both conventional and new cyclones. Efficiency decreases if the cyclone length is above or below the optimum length.
8. The cyclone length which allows maximum collection efficiency increases depending on the increase in the flow rate or in the inlet velocity.
9. Comparison of the results shows that the difference between the pressure drops of the new and classical cyclone with the same size is not significant. Therefore, the decrease of the frictional surface of the new cyclone might increase the intensity and length of the vortex rather than increase the pressure drop. Collection efficiency of the new cyclone is higher than that of the conventional cyclone.
10. The fractional efficiency curves of the new cyclone are steeper than that of the conventional cyclone. The steepness of the curves is changing as the position of the vortex limiter varies. There exists an optimum position of the vortex limiter for the best fractional efficiency curve.
11. The new cyclone design makes possible to control and increase the vortex length and collection efficiency in miniature or small cyclones.

The effects of other geometrical parameters on the performance of the new cyclone and comparison with conventional cyclones should also be investigated in future.

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